

TYLER ROBERTS

Senior Software Engineer

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SUMMARY

Senior Software Engineer with 10+ years building customer-facing **SaaS**, workflow automation, and data-driven business platforms using **React 16–18**, **Next.js 12–14**, **Node.js 14–20**, and **TypeScript 4–5** across product engineering environments where reliability and delivery speed both matter. Strong track record designing scalable frontend architecture, API-first backend services, and event-aware integrations, while modernizing legacy modules, fixing production defects, and shipping measurable UX and performance gains. Known for turning ambiguous product goals into clean releases with strong observability, secure access control, and maintainable code.

SKILLS

- **Frontend** — **React 16–18**, **Next.js 12–14**, JavaScript ES6+, **TypeScript 4–5**, Redux Toolkit, React Query, Context API, HTML5, CSS3, Tailwind CSS, Bootstrap, responsive UI architecture, component libraries, accessibility, SPA routing, form handling, performance optimization
- **Backend & APIs** — **Node.js 14–20**, Express.js, NestJS, REST APIs, GraphQL, microservices, API versioning, authentication, authorization, background jobs, webhook integrations, file processing, validation pipelines, rate limiting, idempotency, caching strategies
- **Data, Messaging & Search** — PostgreSQL, MySQL, SQL Server, Redis, MongoDB, DynamoDB, Kafka, RabbitMQ, AWS SQS/SNS, Elasticsearch, OpenSearch, Redshift, BigQuery
- **Cloud & DevOps** — AWS, Azure, GCP, Docker, Kubernetes, EKS, ECS, EC2, Lambda, API Gateway, S3, CloudFront, RDS, DynamoDB, ElastiCache, EventBridge, Step Functions, Terraform, CloudFormation, GitHub Actions, Jenkins, CI/CD pipelines
- **Observability, Quality & Security** — OpenTelemetry, Datadog, Prometheus, Grafana, CloudWatch, ELK/EFK, Jest, Mocha, Chai, React Testing Library, Cypress, Playwright, Pact-style API testing, OAuth2, OpenID Connect, JWT, RBAC, ABAC, audit logging, OWASP-aligned secure coding

EXPERIENCE

Senior Software Engineer

Web Venturez LLC

February 2020 - October 2025

- Led development of multi-tenant product features using **React 17–18**, **Next.js 13–14**, **Node.js 18–20**, and **TypeScript 4–5**, delivering scalable customer workflows for subscription, reporting, and partner-facing operations in a fast-moving **SaaS** environment.
- Designed reusable frontend architecture with shared hooks, typed service layers, and modular UI patterns that reduced duplicate implementation effort by **35%** and made new feature delivery more predictable across parallel product streams.
- Built and maintained backend services with **Node.js**, Express, and NestJS, exposing stable REST and GraphQL contracts that supported dashboard views, admin tooling, third-party integrations, and audit-sensitive operational actions.
- Resolved a recurring production issue where stale cache entries caused users to see outdated account states by redesigning Redis invalidation rules and event-triggered refresh logic, which reduced support complaints by **40%**.
- Owned migration of older React class-based screens into modern functional components with hooks and stricter **TypeScript** coverage, improving maintainability while lowering frontend regression volume during major release cycles.
- Implemented role-aware access controls and token validation flows using **OAuth2**, **JWT**, and **RBAC**, ensuring sensitive operational screens and administrative actions were consistently protected across web and API layers.
- Introduced server-side rendering and route-level code splitting in key customer journeys, improving first-load performance by **30%+** and materially reducing abandonment on high-traffic onboarding and billing pages.
- Fixed a difficult checkout defect where asynchronous pricing updates intermittently produced mismatched totals by centralizing pricing state, adding backend validation, and introducing defensive contract tests across service boundaries.

- Built background job pipelines for exports, notifications, and reconciliation tasks using queues and retry-safe handlers, preventing duplicate processing during transient infrastructure failures and improving operational trust.
- Improved observability through structured logging, **OpenTelemetry** tracing, and dashboarding in Datadog and Grafana, which made production bottlenecks easier to isolate during high-latency incidents and release verification.
- Partnered with product and design teams to implement new reporting, workflow automation, and customer self-service features, translating incomplete requirements into phased delivery plans with practical technical guardrails.
- Mentored engineers on component design, API consistency, debugging strategies, and migration planning, helping the team adopt cleaner release practices while keeping velocity steady through a period of active platform modernization.

Software Engineer

Webiz Digital

July 2017 - January 2020

- Delivered full-stack business applications using **React 16–17**, JavaScript, **Node.js 14–16**, Express, and SQL-backed services, supporting internal operations and customer-facing modules for web-first product teams.
- Built dynamic dashboard pages, admin panels, and data-entry workflows with reusable React components, improving consistency across views and reducing UI implementation time for new requests by **25%**.
- Developed backend APIs for account management, content flows, and transactional operations, with careful validation and error handling that reduced malformed request failures and improved downstream service stability.
- Contributed to migration of legacy jQuery-driven pages into component-based **React** interfaces, enabling more maintainable state handling and smoother rollout of modern UX capabilities without forcing risky full rewrites.
- Diagnosed and fixed a persistent bug where duplicate form submissions created repeated records under slow network conditions, solving it through button state locking, idempotent API handling, and clearer user feedback.
- Enhanced search and filtering experiences by integrating **Elasticsearch** and optimized SQL querying patterns, making large operational datasets easier to navigate while lowering page response times by **20%**.
- Implemented file import and export tooling for operational users, including validation summaries and recoverable error messaging, which reduced failed uploads and gave support teams far better troubleshooting context.
- Containerized local and test environments with **Docker** and improved CI flows in Jenkins and GitHub Actions, reducing environment drift and helping releases move through verification with fewer deployment-related surprises.
- Worked on incremental backend refactoring that moved monolithic service logic into cleaner Express modules and shared utilities, improving testability and making cross-team ownership less fragile during active product changes.
- Introduced caching, pagination, and query tuning strategies across heavy endpoints, which improved perceived responsiveness and lowered database pressure during peak usage windows by roughly **30%**.
- Collaborated closely with stakeholders to implement new customer portal features, billing support enhancements, and admin workflow improvements, balancing quick delivery needs with long-term code health and operational resilience.

Software Developer

Centric Tech Inc.

October 2013 - June 2017

- Developed internal and client-facing web applications using JavaScript, server-rendered frontend patterns, **Node.js 10–12**, Express, and relational databases, building a strong foundation in reliable business software delivery.
- Created CRUD-heavy workflow screens, reporting modules, and approval paths that supported day-to-day operational teams, with careful backend validation and SQL tuning to keep data integrity strong under frequent use.

- Supported early modernization efforts by introducing modular JavaScript components and API-backed interactions into older pages, improving maintainability while respecting the constraints of stable legacy business systems.
- Resolved a recurring concurrency problem where overlapping edits silently overwrote records, solving it with version checks, clearer validation feedback, and safer update handling that improved user trust in shared workflows.
- Improved import processing jobs by adding stronger validation, retry logic, and detailed error reporting, which reduced failed batch reruns and gave support engineers clearer insight into malformed operational data.
- Optimized slow database-backed screens through query analysis, indexing changes, and reduced server round-trips, producing noticeable load-time gains and improving the usability of high-volume internal modules.
- Implemented authentication and permission checks across administrative tools, ensuring sensitive functions were available only to the right user groups and reducing security-related review findings in release testing.
- Worked with QA and support teams to trace production bugs from user reports into logs, SQL traces, and backend handlers, developing a disciplined debugging approach that consistently converted vague issues into permanent fixes.
- Assisted with gradual service extraction and API standardization efforts that prepared older modules for later platform upgrades, making future migration work less risky and more predictable for the wider engineering team.
- Delivered practical feature enhancements for reporting, notifications, and operational dashboards while staying flexible across frontend, backend, and database work, which made the team more effective during resource-constrained delivery cycles.

EDUCATION

Bachelor's degree in Computer Science

Florida State University

2009 – 2013

- Built a strong foundation in software engineering, databases, algorithms, and application design that supported later work across **React**, **Node.js**, cloud systems, and production web architecture.